Damian **Sowinski**, Ph.D.

Theoretical Physicist | Complex Systems, Information Theory & Artificial Intelligence

Contact

DRSowinski@gmail.com

Webpage

DamianSowinski.com

Languages

English *native*
Polish *fluent*

Programming

MATLAB, Python
Mathematica,
Fortran
C, C++

External Links

Webpage
arXiv
ORCID
Lectures
Wordpress
Visualizations
Google Scholar

ToC

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Comm. 5-8
Pub. 9-11

Summary

Theoretical and computational physicist working at the intersection of complex systems, information theory, and artificial intelligence. My research develops physics-based frameworks to study fundamental questions about agency, collective behavior, and the emergence of life — drawing on tools ranging from neuroevolution and machine learning to stochastic simulation and continuum mechanics. I maintain an active publication record in journals including Physical Review, Nature Communications Physics, and PRX Life, and hold active collaborations with groups at Rochester, Delaware, Sherbrooke, and VUB in Brussels, with grant proposals pending at Templeton, Sloan, and NASA. My work is inherently cross-disciplinary, and I have a strong record of building collaborative research programs and mentoring students at multiple levels.

Education

Doctor of Philosophy**, Physics** — Dartmouth College, New Hampshire **2016**

Bachelor of the Arts**, Physics** — University of California, Berkeley **2003**

Research

Department of Physics and Astronomy, University of Rochester — Rochester, New York **2023-Present**

Postdoctoral Researcher/Adjunct Professor

Collaborate with **Adam Frank** and **Gourab Ghoshal** on understanding **information storage, transmission, and processing** in biological, ecological, and societal settings. This work is part of a collaboration with **Artemy Kolchinsky** and **Jordi Pinero** at Pompeu Fabra University under a three-year **Templeton Foundation** Grant, with the goal of operationalizing a theory of **semantic information** using techniques from **stochastic thermodynamics**. Administrative responsibilities include leading **weekly meetings,** organizing a **workshop** at the University of Rochester in 2024, running a **weekly seminar on General Relativity**, and **advising multiple graduate and undergraduate students.** Collaborate with **Matthew Egbert** on **agent-based simulations** of foraging, investigating the informational aspects of both internal and external sensors. Work with **Dora Biro** and **Takao Sasaki** on an interdisciplinary project to map the **information architecture** of **Solenopsis Invicta** during foraging behavior. Develop models of **learning** in **homing pigeon flights** and created the **Layered Agent to Environment Foraging Simulation (LEAFS) framework** for modeling the emergence of **collective problem solving** by colonies.Founded the **Complex Systems and Integrated Information** collaboration in 2023, with responsibility for scheduling speakers and administrative coordination. Collaborate with **Stuart Bartlett, Sylvia Holler, Henderson Cleaves, Mike Wong**, and others on applying **computational mechanics** to pyrolysis data and **exoplanet spectra**. Participate in **Clement Vidal's SETI collaboration** on **technosignatures.**

Thayer School of Engineering, Dartmouth College — Hanover, New Hampshire **2019-2022**

Research Scientist

Collaborate with **Matthew McGarry** and **Elijah Van Houten** on developing a **parallel finite element computational suite** for both **forward and inverse simulations of continuum systems**, focusing primarily on **poroelastic** and **anisotropic Neo-Hookean continuum models** with applications toward **imaging tissue properties**. Assist with **curriculum development** for **Keith Paulsen's Finite Element Course**. Work with **Scott Gordon** on building **Raspberry Pi integrated sensors** for **rheological tests** of multiple porous biological and non-biological materials. Train at the **Center for Surgical Imaging** at **Dartmouth-Hitchcock Medical Center** to run **MRI sequences** on phantoms. Contribute to **grant proposal writing** for **diabetic foot studies** and create a **finite strain adipose-septae model** for foot pad tissue. Collaborate with **Richard Granger** and **Elijah Bowen** in the **Psychology and Brain Sciences** department on a **Riemannian metric** inspired by the **cellular architecture of the eye** for **image similarity judgments**, as well as **differential geometry** approaches toward the **Weber-Fechner law.** Work with **Daniel Reeves** at the **Fred Hutchinson Cancer Research Center** on a **field theory analogue** of **ecological models** applied to interacting systems of heterotrophs and autotrophs. Collaborate with **Marcelo Gleiser, Jeremy DeSilva, Jonathan Carroll-Nellenback**, and others on modeling **information dynamics in groups** of epistemic agents.

Research (Continued...)

Institute for Cross-Disciplinary Engagement, Dartmouth College Hanover, New Hampshire **2017-2019**

Postdoctoral Scholar

Build the theoretical foundation of **Configurational Information Measures (CIM)** and explore its phenomenology in physical systems with **Marcelo Gleiser.** Develop **pyCE, a Python library** for working with CIMs that also simulates and analyzes **astrophysical** and **cosmological phenomena**, maintaining and optimizing the codebase. Work with graduate students **Michelle Stephens** and **Max Krakow** to ensure continuity of the code base and analysis techniques. Collaborate with **Seth Frey** at **UC Davis** on **data processing and information theoretic analysis** of **NBA video** using the **MaxEnt** method. Work with **Shirley Ho** at the Center for **Computational Astrophysics (CCA)** on **machine learning** approaches for **21cm measurements in cosmology**. Collaborate with **Sonali Mohapatra, Chen Sun**, and **Djuna Croon** on a **Boson star-dark matter** model using **numerical relativity** and finite **difference methods**.

WisPIC, University of Wisconsin Madison, Wisconsin **2016-2017**

Postdoctoral Researcher

Analyze **information measures** applied to **Ising models on graphs** representing distinct **neurotopologies**. Write and maintain a **statistical mechanics pipeline** for **data generation and analysis**. Work closely with **Sabrina Streipert** on the **split brain phenomenon** and methods for modeling it using both **deterministic and stochastic networks**. Apply **Ising, Potts, and Heisenberg models** on both **lattices and Erdős–Rényi graphs**. Collaborate with **Leonardo Barbossa** on analyzing the statistical properties of **spin glass systems** used to describe **complex network dynamics**.

Beyond Institute, University of Arizona Tempe, Arizona **2015**

Visiting Scholar

Collaborate with **Sara Imari Walker**, **Hyunju Kim**, and **Alyssa Adams** on **information processes** in biological systems. Develop a model of **stochastic cellular automata** that combines both deterministic dynamics and Ising-like behavior. Analyze the competition between deterministic and stochastic subprocesses in the automata. Work on a **machine learning approach** for understanding the **yeast cell growth cycle**.

Department of Physics & Astronomy, Dartmouth College Hanover, New Hampshire **2011-2016**

Graduate Student

Develop the use of **Configurational entropy** in field theory under the guidance of **Marcelo Gleiser**. Work on understanding entropic measures relationships to physical systems via an **informational narrative.** Develop code to **simulate Landau-Ginzburg models** on a lattice and extract critical exponents. Learn to write and optimize code in both **MATLAB** and **Python**, master **numerical methods**, and become an expert in **finite difference schemes** as applied to field theory and classical physics.

Hayden Planetarium, American Museum of Natural History New York, New York **2010-2011**

Research Assistant

Work under **Mordecai Mark Mac-Low** and **Colin McNally** on a modified **GADGET code** to understand the numerically induced viscosity in **hydrodynamic simulations**.Gain extensive experience with **C programming**.

Department of Astronomy, Hunter College New York, New York **2010-2011**

Research Assistant

Write **Python** code to read and analyze the **spectra of M and L dwarf stars**, characterizing the populations in color space for future photometric surveys by the **Large Synoptic Survey Telescope (LSST)**.

Lawrence Berkeley National Laboratory Berkeley, California **2000-2003**

Undergraduate Researcher

Work for **Herbert Steiner** in the **KamLAND** collaboration to develop the z-axis deployment arm for detector calibration. Perform **sensitivity analysis** on **photomultiplier tubes**.

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Teaching/Advising

University of Rochester Rochester, New York **2025-Present**

Adjunct Professor

Teaching and developing the **Physics 100: Physics and the Natural World** course for non-STEM majors. Taught **Physics 141: Honors Mechanics** for incoming freshmen hoping to major in physics. Developed a curriculum around **Kleppner & Kolenkow**. Developing the curriculum for all the non-honors physics labs, including organization of lab materials, instructor setup instructions, and student lab manuals.

University of Rochester Rochester, New York **2023-Present**

Advising/Mentoring

Advise **graduate student Sean Kelty** on his work on **information dynamics** in the context of the Ising Model, applying configurational entropy measures to identify critical points with minimal scaling analysis. Advise **graduate student Binod Bhatt** on his first research project involving **entropy estimation** for analyzing functional MRI data. Advise **graduate student Bobbie Markwick** on research simulations using my foraging model. Lead an **advanced topics seminar** titled **"My Favorite Spacetimes"** for graduate and advanced undergraduate students, focusing on specific solutions to **General Relativity**. The first three seminars cover the **history** of the **luminiferous aether** and **Special Relativity, Minkowski spacetime**, and **Rindler spacetime**. Remaining seminars consist of **reading classical papers** on particular spacetime solutions and working out well-known results to give students broad perspective on the subject. Cover both cosmological solutions (**FRW, Gödel, van Stockum**) and stellar solutions (**Schwarzschild, Kerr, Reissner-Nordström, Tolman-Oppenheimer-Volkhoff**).

Thayer School of Engineering Hanover, New Hampshire **2019**

Co-Lecturer

Update the curriculum for **Keith Paulsen's Engineering 105** course on **finite element methods.** Deliver lectures during Dr. Paulsen's travel periods. Maintain **open office hours** to help graduate students with troubleshooting **FORTRAN code**.

RIA Global Education Online Course **2019**

Lecturer

Develop curriculum for and teach a **Physics of Music** online course at the **undergraduate level**. Contribute to coursework development on General Relativity, Cosmology, and Black Holes.

Dartmouth College Hanover, New Hampshire **2011-2016**

Graduate Student Teaching Assistant

Assist teaching **Honors Introductory Physics courses** and build rapport with incoming undergraduates by **organizing extracurricular mathematics and programming sessions**. Serve as TA for courses including classical mechanics, electromagnetism, introductory astronomy, introductory quantum mechanics, advanced quantum mechanics, special and general relativity, cosmology, and quantum field theory.

Hunter College New York, New York **2010-2011**

Educational Consultant

Develop a **Physics GRE test prep course** for the **physics department** and organize tutoring sessions for undergraduates. Teach an introduction to Python course.

Beginning with Children Foundation New York, New York **2008-2011**

Educational Consultant

Develop curriculum for **after-school tutoring** and **test prep programs**, focusing on **mathematics** and the **SSAT**, **SHSAT**, and **SAT** exams. **Mentor students** with academic habits and college applications.

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Teaching/Advising (Continued...)

Launch Education New York, New York **2008-2011**

Tutor

Work with students on **high school level biology** and **mathematics**.

Tutor Associates New York, New York **2005-2011**

Head Physics and Mathematics Tutor

Work with students on **AP physics** and **AP calculus** material, as well as **test prep**. Develop **dynamic spreadsheets** for the company to **analyze student performance** over time.

Celia Cruz Bronx High School of Music New York, New York **2005-2007**

Physics Teacher

Develop the curriculum for and teach the entire spectrum of **physics courses**. Conduct **statistical analysis on student body performance** for the **NYC Department of Education**. Achieve **higher than city average passing rates on the Regents Exams**.

Grace Dodge High School New York, New York **2006**

Mathematics Teacher

Teach the **Math A course** during the summer. Higher than average passing rates on the Regents Exam.

UC Tutors Irvine, California **2004-2005**

Tutor

Work with undergraduates on **physics** and **single/multi variable calculus** coursework. **Math GRE** prep.

SmArT Academy Cerritos, California **2003-2005**

Physics and Mathematics Teacher

Teach physics, **AP physics,** and **AP calculus** courses. **Achieve 100% passing rate on the AP Physics exam**.

Fullerton Community College Cerritos, California **2003-2005**

Head Science and Mathematics Tutor

Work with college students on **introductory chemistry, physics**, **statistics**, and **calculus** courses in the school's **tutoring center**.

Cragmont Elementary School Berkeley, California **2001-2003**

Work Study Teacher's Assistant

Work with **elementary school** children in 1st-4th grade on **basic arithmetic skills**.

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Communications

Laplace's Demon and the Drunkard — **Autumn 2025**
Complexity Collaboration, Zoom Talk

Cosmic Strings and the CP Problem — **Autumn 2025**
University of Rochester, Department of Physics and Astronomy, Journal Club Talk

Time Machines and Warp Drives: General Relativity and Sci-fi — **Summer 2025**
McGregor's Tavern, Nerd Night Public Lecture

Information Dynamics in Paired Homing Pigeon Flights — **Spring 2025**
University of Rochester, Department of Physics & Astronomy, Group Talk

What is Complexity? Perspectives from Computational Mechanics — **Spring 2025**
University of Rochester, Department of Physics & Astronomy, Seminar

Top-Down Bottom-Up Causation in Collective Behavior — **Winter 2025**
University of Rochester, Animal Behavior and Cognitive Science, Seminar

Time Machines, Wormholes, and Warp Drives — **Autumn 2024**
University of Rochester, Department of Physics & Astronomy, Department Talk

My Favorite Spacetimes — **Autumn 2024**
University of Rochester, Department of Physics & Astronomy, 10 week seminar

Why Can't We Agree? — **Summer 2024**
University of Rochester, Animal Behavior and Cognitive Science, Department Talk

The Meaning in Life: Towards a Theory of Semantic Information — **Summer 2024**
University of Rochester, Information Driven States of Matter Workshop, Talk

Information Sharing Polities, Cognitive Architectures, and Consensus — **Summer 2024**
University of Rochester, Complex Systems and Information Theory Meeting, Talk

Biospheres and Planets: Semantic Information Theory & the Gaia Hypothesis — **Spring 2024**
Astrobiology Science Conference 2024, Poster

Emergence of Meaning in the Sensor Evolution of Ante-organisms — **Spring 2024**
American Physical Society, March Meeting, Talk

Much Ado About Nothing: Vacuum Phase Transitions & Topological Defects — **Spring 2024**
University of Rochester, Astronomy Journal Club, Talk

Consensus in Polities with Heterogeneous Cognitive Architectures is Hard — **Winter 2024**
Northeast Conference on Complex Systems 2024, Talk

Semantic Information in Agent-Based Foraging Models. — **Autumn 2023**
University of Rochester, Complex Systems and Information Theory Meeting, Talk

Naked Singularities in Critical Gravitational Collapse — **Autumn 2023**
University of Rochester, Astronomy Journal Club, Talk

Semantic Information in the Kuramoto Model — **Summer 2023**
University of Rochester, Landi's Quantum Systems Group, Seminar

Multi-Timescale Nanoparticle Simulations — **Summer 2023**
Dartmouth College, Nanoparticle Imaging Group at Thayer School of Engineering, Zoom Talk

Finding Meaning in Finding Food — **Winter 2023**
Northeast Conference on Complex Systems 2023, Talk

Semantic Information in Foraging Models — **Autumn 2022**
University of Rochester, Zoom Seminar

Simulating Nanoparticle Dynamics II — **Summer 2022**
Dartmouth College, Nanoparticle Imaging Group at Thayer School of Engineering Talk

Simulating Nanoparticle Dynamics I — **Summer 2022**
Dartmouth College, Nanoparticle Imaging Group at Thayer School of Engineering Talk

Numerical Techniques for Stochastic Differential Equations — **Spring 2022**
Dartmouth College, Nanoparticle Imaging Group at Thayer School of Engineering Talk

Communications (Continued...)

Lie Algebras and Conformal Transformations in Phase Transitions **Winter 2022**
Collaboration with Zajj Daugherty at the City College, NYC

Information Transfer in Ising Models with Temperature Gradients **Winter 2022**
University of Rochester, Psychohistory Collaboration Zoom Talk

Renormalization Group Flow and the Acoustoelastic Effect in Gel Phantoms **Autumn 2021**
Dartmouth College, MRE Group at Thayer School of Engineering Zoom Talk

Langevinference - a novel approach to property inference in MRE **Autumn 2021**
Dartmouth College, MRE Group at Thayer School of Engineering Zoom Talk

The Acoustoelastic Effect and Hyperelasticity **Summer 2021**
Dartmouth College, MRE Group at Thayer School of Engineering Zoom Talk

Information Geometry and Entropic Measures in Data Analysis **Spring 2021**
Dartmouth College, Psych and Brain Science Department Talk

Hydraulic Permeability Reconstruction in Magnetic Resonance Elastography **Spring 2021**
American Physical Society, March Meeting, Talk

Agent Based Predator-Prey Flocking Models **Spring 2021**
University of Rochester, Psychohistory Collaboration Zoom Talk

Criticality in the Presence of Physical Gradients **Spring 2021**
University of Rochester, Psychohistory Collaboration Zoom Talk

Coarse Graining in Lotka-Volterra Systems **Winter 2021**
Dartmouth College, Ecological Dynamics Collaboration Zoom Talk

The Irreducible Representation Expansion of Hyperelasticity **Winter 2021**
Dartmouth College, Anisotropic MRE Collaboration Talk

Modeling Networks of Interacting Species **Winter 2020**
Dartmouth College, Ecological Dynamics Collaboration Zoom Talk

Diabetic Foot - Biomechanical Aspects of Finite Deformations **Autumn 2020**
Dartmouth College, MRE Group at Thayer School of Engineering Zoom Talk

A Survey of Biomechanical Models **Autumn 2020**
Dartmouth College, MRE Group at Thayer School of Engineering Zoom Talk

Biomechanics and Arterial Modeling **Autumn 2020**
Dartmouth College, MRE Group at Thayer School of Engineering Zoom Talk

The Foundations of an Effective Field Theory of Tissue **Summer 2020**
Dartmouth College, Thayer School of Engineering 4 part Lecture Series

The Continuum Mechanics of Infinitesimal and Finite Deformations **Spring 2020**
Dartmouth College, Thayer School of Engineering 2 part Lecture Series

Regularization Methods in MRE **Autumn 2019**
BioMedical Engineering Society Annual Meeting

A Bayesian Perspective on Optimization **Summer 2019**
MRE internal annual collaboration meeting

Information, Quantum Mechanics, and the Eyeball **Summer 2019**
Dartmouth College, Psych and Brain Science Department Talk

Information Content in Natural Images **Summer 2019**
Dartmouth College, Psych and Brain Science Department Talk

Logit and Probit Functions for Classification in Psychometric Analysis **Spring 2019**
Dartmouth College, Psych and Brain Science Department Talk

Vector Spaces and Mathematical Methods for Psychophysics **Spring 2019**
Dartmouth College, Psych and Brain Science 4 part Lecture Series

Communications (Continued...)

The Weber-Fechner Law and Perception Metrics — **Winter 2019**
Dartmouth College, Psych and Brain Science Department Talk

Do you even Latex? Part II — **Autumn 2018**
Kresge Library at Dartmouth College, 4 session Workshop

The ADM Decomposition and Numerical relativity — **Autumn 2018**
Dartmouth College, Physics Department Talk

Oscillons in Vacuum Phase Transitions — **Autumn 2018**
American Physical Society, New England Meeting, Poster

Do you even LaTeX? — **Autumn 2018**
Kresge Library at Dartmouth College, 4 session Workshop

Finding Boson Stars Numerically — **Summer 2018**
Dartmouth College, Physics Department Talk

pyCE: A Python Library for Configurational Entropy — **Summer 2018**
Dartmouth College, Physics Department Talk

Steaming Piles of Science: Dr. Sowinski on Information Theory — **Winter 2018**
Interview for a popular science podcast (link)

An Ode to Oscillons: Criticality in QFT — **Winter 2018**
McGill Northeast Cosmology Workshop, Talk

A Minimal History of Maximal Entropy — **Winter 2018**
Dartmouth College, Postdoc Association Public Lecture

MIB Coordinates and Finite Difference Methods for QFT Simulations — **Autumn 2017**
Dartmouth College, 2 Part Lecture Series

Criticality and Information Dynamics in Neuroscience — **Spring 2017**
WISPIC, Neuroscience Department Talk

Ising Models in Neuroscience — **Winter 2017**
WISPIC, Neuroscience Department Talk

From Epistemology to Information — **Autumn 2016**
Institute for Cross-Disciplinary Engagement, 3 part Lecture Series

MIB Coordinates and Finite Difference Methods for QFT Simulations — **Autumn 2017**
Dartmouth College, 2 Part Lecture Series

Criticality and Information Dynamics in Neuroscience — **Spring 2017**
WISPIC, Neuroscience Department Talk

Ising Models in Neuroscience — **Winter 2017**
WISPIC, Neuroscience Department Talk

From Epistemology to Information — **Autumn 2016**
Institute for Cross-Disciplinary Engagement, 3 part Lecture Series

The Epistemic Foundations and Phenomenology of Configurational Entropy — **Summer 2016**
Dartmouth College, Physics Department Lecture

Information Turbulence at Criticality — **Spring 2015**
Carnegie Institute of Science, Poster Session

Maximum Entropy Methods in Statistical Physics — **Winter 2015**
Dartmouth College, 2 part Lecture Series

Probability, Information, and Statistical Physics. — **Autumn 2014**
Dartmouth College, 4 part Lecture Series

Communications (Continued...)

The Cox Axioms: A Foundational Approach to Bayesian Probability Theory — **Summer 2014**
Dartmouth College, 2 part Lecture Series

Geometry, Topology, and Physics by Nakahara — **Spring 2014**
Dartmouth College, Reading Course

Configurational Entropy and the Chandrasekhar Limit — **Autumn 2013**
American Physical Society April Meeting, Talk

Cosmic Anisotropy: A History of the Axis of Evil — **Spring 2013**
Dartmouth College, Physics Department Talk

A Multi-Letter Polymerization Model of Chiral Symmetry Breaking — **Summer 2012**
Astrobiology Graduate Student Conference, Poster Session

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Publications (click to link)

Behavioral Speciation in the Neuroevolution of Recurrent Neural Net Foragers driven by Environmental Homogeneity
Damian R. Sowinski
Submitted to *Artificial Life Conference 2026*

Causal Leverage Density: A Universal Framework for Semantic Information
Damian R. Sowinski, Silvia Holler, Mike Wong, Gary Segal, David Parkinson, Clément Vidal, James Henderson Cleaves, Pritvik Sinhadc, Anirudh Prabhu, Stuart Bartlett
Artificial Life Conference Proceedings 37. Vol. 2025. No. 1, MIT Press.

Notions of adiabatic drift in the quantized Harper model
Alice C Quillen, Nathan Skerrett, Damian R. Sowinski, Abobakar S. Miakhel
Physical Review A 112 (4), 042226

Exo-Daisy World: Revisiting Gaia Theory through an Informational Architecture Perspective
Damian R Sowinski, Adam Frank, Gourab Ghoshal
The Planetary Science Journal 6.7 (2025): 176.

Information Bounds Production in Replicator Systems
Jordi Piñero, Damian R Sowinski, Gourab Ghoshal, Adam Frank, Artemy Kolchinsky
Accepted, Nature Communications Physics

Reflections of Life: Distinguishing Living from Non-living Worlds with Complexity Metrics
Stuart Bartlett, Xianlej San, Siteng Fan, Gary Segal, Damian R Sowinski, Yuk L Yung
Under Review, The Astrophysical Journal

In vivo Magnetic Resonance Imaging of the Interstitial Pressure Gradients using a Pulsatile Poroelastic Computational Model
Matthew McGarry, Damian Sowinski, Likun Tan, Elijah Van Houten, Jaco J.M. Zwanenburg, John Weaver, Keith Paulsen
Interface focus 15 (1), 20240044

Forager Population Viability in Varying Environments
Robert Marwick, Damian Sowinski, Adam Frank, Gourab Ghoshal
Working Paper

Configurational Information Measures, Phase Transitions, and an upper bound on Complexity
Damian Sowinski, Sean Kealty, Adam Frank, Gourab Ghoshal
Under Review, Physical Review E

Hierarchical Fiber Model Building for Magnetic Resonance Elastography
Damian Sowinski, Matthew McGarry, Curtis L Johnson
Working Paper

Semantic Information in a feedback-control Kuramoto Model
Damian Sowinski, Adam Frank, Gourab Ghoshal
Phys. Rev. Research 6, 043188 2024

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Publications (click to link)

Semantic Information in a Model of Resource Gathering Agents
Damian Sowinski, Jonathan Carroll-Nellenback, Jordi Pinero, Artemy Kolchinsky, Robert Marwick, Adam Frank, Gourab Ghoshal, Marcelo Gleiser
PRX Life 1 (2), 2024

The Consensus Problem in Polities of Agents with Dissimilar Cognitive Architectures
Damian Sowinski, Jonathan Carroll-Nellenback, Jeremy M DeSilva, Adam Frank, Gourab Ghoshal, Marcelo Gleiser
Entropy 24 (10), 1378

Visual Stream Connectivity Predicts Assessments of Image Quality
Elijah Bowen, Antonio Rodriguez, Damian Sowinski, and Richard Granger
Journal of Vision 22 (11), 4-4

Consensus Between Epistemic Agents is Difficult
Damian Sowinski, Jonathan Carroll-Nellenback, Jeremy M DeSilva, Adam Frank, Gourab Ghoshal, Marcelo Gleiser, Hari Seldon
arXiv:2201.04642

Mapping Heterogenous Anisotropic Tissue Mechanical Properties with Transverse Isotropic Nonlinear Inversion MR Elastography
Matthew McGarry, Elijah Van Houten, Damian Sowinski, Dhrubo Jyoti, Daniel Smith, Diego A Caban-Rivera, Grace McIlvain, Philip Bayly, Curtis L Johnson, John Weaver, Keith Paulsen
Magnetic Resonance in Medicine 87 (3), 1313-1328

Quantifying Stability of Parameter Estimates for In Vivo Nearly Incompressible Transversely-Isotropic Brain MR Elastography
Dhrubo Jyoti, Matthew McGarry, Elijah Van Houten, Damian Sowinski, Philip V Bayly, Curtis L Johnson, Keith Paulsen
Biomedical Physics & Engineering Express 8 (3), 035015

Poroelasticity as a Model of Soft Tissue Structure: Hydraulic Permeability Reconstruction for Magnetic Resonance Elastography In Silico
Damian Sowinski, Matthew McGarry, Scott Gordon-Wylie, Elijah Van Houten, John Weaver, and Keith Paulsen
Frontiers of Physics, Research Topic: Innovative Dev. in Multi-Modality Elastography (2020)

Correlated Noise in Brain Magnetic Resonance Elastography
Ariel J. Hannum, Grace McIlvain, Damian Sowinski, Matthew D. J. McGarry, Curtis L. Johnson
Magnetic Resonance in Medicine (2021) 00: 000– 000.

Towards a Perceptual Distance Metric for Auditory Stimuli
Sarah Oh, Elijah Bowen, Antonio Rodriguez, Damian Sowinski, Eva Childers, Annemarie Brown, Laura Ray, and Richard Granger
arXiv:2011.00088

A Heterogenous, Time Harmonic, Nearly Incompressible Transverse Isotropic Finite Element Brain Simulation Platform for MR Elastography
Matthew D J McGarry, Elijah Van Houten, Charlotte Guertler, Ruth J Okamoto, Daniel R Smith, Damian R Sowinski, Curtis L Johnson, Philip Bayly, John Weaver, and Keith D Paulsen
Physics in Medicine & Biology (2020)

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Publications (click to link)

Configurational Information Approach to Instantons and False Vacuum Decay in D-dimensional Spacetime
Damian Sowinski , Marcelo Gleiser
Physical Review D, 98, 056026 (2018)

Configurational Entropy as a Lifetime Predictor and Pattern Discriminator for Oscillons
Marcelo Gleiser, Michelle Stephens , and Damian Sowinski
Physical Review D 97, 096007 (2018)

How We Make Sense of the World: Information, Map-Making, and the Scientific Narrative
Marcelo Gleiser, Damian Sowinski
The Map and the Territory: Exploring the Foundations of Science, Thought and Reality, ed. by S. Wuppuluri, N. da Costa, and F. A. Doria, Springer Frontiers Collection (2018)

Information Dynamics at a Phase Transition
Damian Sowinski , Marcelo Gleiser
Journal of Statistical Physics, Volume 167, Issue 5, pages 1221-1232, June 2017

Complexity and Stability for Epistemic Agents: The Foundations and Phenomenology of Configurational Entropy
Damian R. Sowinski
Dartmouth College, Department of Physics and Astronomy, Thesis (2016)

Information-Entropic Signature of the Critical Point
Marcelo Gleiser, Damian Sowinski
Physics Letters B, Volume 747, Pages 125-128, 30 July 2015

Information-Entropic Stability Bound for Compact Objects: Application to Q-Balls and the Chandrasekhar limit of Polytropes
Marcelo Gleiser, Damian Sowinski
Physics Letters B, Volume 727, Issues 1-3, Pages 272-275, 25 November 2013

Cosmic Anisotropy: A History of the Axis of Evil
Damian Sowinski
Dartmouth College, Scribd article 130092552 (2013)

From Monopoles to Textures: A Survey of Topological Defects in Cosmological Quantum Field Theory
Damian Sowinski
Dartmouth College, Scribd article 85012149 (2012)

Photometric Analysis of M&L Dwarfs with the LSST Throughput Profiles
Damian Sowinski
American Museum of Natural History, Scribd article 35701963 (2010)